

SHWETA MISHRA

Boston, MA • (857) 260-9071 • mishra.b.shw@gmail.com • [linkedin.com/in/shwm](https://www.linkedin.com/in/shwm) • github.com/MishraCo

Portfolio: mishraco.github.io/ShwetaPortfolioWebsite/

SUMMARY

Seasoned ML Engineer with 3+ years of experience in Python-based data science. Expert in transforming intricate requirements into tangible, data-centric solutions. Demonstrated success in Agile project delivery and adept at engaging with executive stakeholders. Eager to apply data analytics acumen and ML/AI proficiency to foster data-driven decision-making and innovative solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, PostgreSQL, R, C++, Java, HTML, CSS, MERN Stack.
- **Libraries and Frameworks:** Pandas, NumPy, Statsmodel, Scipy, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Plotly, Seaborn, OpenCV, NLTK, Spacey, Beautiful Soup, Databricks Spark, MLlib, MS Excel.
- **Machine/Deep Learning:** Cross Validation, OLS/GLM/Regression, Logistic Regression, SVM, Decision Trees, **Random Forest**, XGBoost, CatBoost, K-Means, Collaborative Filtering, CNN, Transformer, NLP, CV.
- **Data Management:** Power BI, Tableau, Looker, AWS QuickSight, Hadoop, Apache Spark SQL, Databricks.
- **Database:** Oracle (GL/Financial data/ FSCM), SAP, Azure, Redshift, Snowflake, Hive, MongoDB.
- **Software Development:** Version Control (**Git**), Jenkins, Jira, Docker, Kubernetes, Airflow, Confluence.
- **Analytical Skills:** Statistical modeling, Hypothesis Testing (T-tests, ANOVA, Chi-Square, A/B testing), and causal inference, Predictive Modeling, Product Analytics, Kafka.
- **Cloud:** AWS (EMR, Redshift, S3, Lambda, Sagemaker), GCP (BigQuery, Compute Engine), MS SQL Server, AzureML, Azure Functions, Snowflake.

PROFESSIONAL EXPERIENCE

TENNECO

Northville, MI

Data Scientist

May 2023-Sep 2023

- Investigated and quantified the seasonality of order patterns, leveraging **advanced analytics** to refine procurement schedules, which minimized emergency orders and associated premium freight costs by 25%.
- **Forecasted** financial outcomes based on a revamped pricing structure using **LSTM** and **ARIMAX**, predicting a 2.5% increase in profit margins, and contributing to a more cost-effective supply chain model.
- Crafted a **Tableau** dashboard using **SQL & Python** to track sales **KPIs** and order trends, driving a 5% profit margin increase and supply chain optimization.
- Designed and published a dynamic **Power BI** dashboard with automated **alerts**, tracking vendors' annual returns credit allowance, and ensured monthly data refresh via **Power Automate**, boosting return management efficiency by 40%.

AUXIAI

Boston, MA

Machine Learning Engineer

Jan 2023-May 2023

- Spearheaded the integration of **ChatGPT's Ada/DaVinci** models to automate content creation for PowerPoint presentations, interpreting user prompts to generate relevant slide material, enhancing content development efficiency by 50%.
- Collaborated with UI/UX teams to integrate **user feedback** into the model training process, which significantly enhanced the model's responsiveness to diverse artistic styles and inputs.
- Pioneered the deployment of an automated **testing** framework within the CI/CD pipeline, which led to a 40% decrease in post-deployment issues and a more robust content generation tool.
- Conducted thorough user training sessions and created comprehensive **confluence documentation**, resulting in a 70% increase in user adoption rate for the new PowerPoint automation tool within the first quarter post-deployment.
- Led a **data augmentation** initiative that expanded the training **dataset** by 50%, incorporating a wider variety of sketch styles and complexities, thereby improving the model's generalization capabilities.

ACCENTURE

Data Scientist

Bengaluru, IND

May 2018-Sep 2021

- Automated **data validation** and ingestion (ETL) of millions of rows using **Spark** for end-of-month and year-end processing, employing **SQL** and **python** scripting, which enhanced data processing efficiency by 70%.
- Formulated a sophisticated driver forecast and simulation framework using **time-series multivariate analysis**, which identified and quantified sales drivers, leading to a 15% increase in the precision of pricing and promotional strategies, and a 10% growth in customer satisfaction related to product offerings.
- Crafted and disseminated **Tableau** based dynamic reports, incorporating a custom scoring system that guided business units in procurement decisions, resulting in a 25% improvement in inventory turnover across key product categories.
- Led & implemented a customer **segmentation** proof of concept using Elbow Method and **K-Means** Clustering, resulting in a 40% uplift in customer experience by enabling the organization to tailor services and facilities to distinct customer groups based on algorithmically determined attributes.
- Designed training and peer-**mentoring** programs for the incoming class of 3 analysts in 2020; reduced onboarding time for new hires by 50%.

PROJECTS

Visual Question & Answering (VQA) - Personal Project | Image Captioning Dec 2023-Jan 2024

- Developed a Streamlit app with **CNN-LSTM** based image content query model utilizing **NLP** and **computer vision** concepts, enhancing interaction with educational content, achieving a 0.66 WUPS score.

Multimodal Forecasting of Solar Irradiance - Academic Project | Transfer Learning Nov 2023-Dec 2023

- Engineered a multimodal forecast system, scoring a 6.6% MAPE and **RMSE**, using **Prophet** and **Inception V4**, showcasing my skills in **time-series forecasting** and **transfer learning** of deep neural networks.

News Text Summarization - Academic Project | NLP, BART, BLEU, ROUGE Score Sept 2023-Oct 2023

- Achieved a BLEU score of 0.85 in news text summarization using GloVe, seq2seq, and T5 models.
- Implemented TF-IDF, document categorization, and feature creation for automatic news text summarization.

Skin Cancer Diagnosis with DL and CV- Academic Project | Transfer Learning, CNNs Jan 2023-Mar 2023

- Detected skin cancer with deep learning and computer vision, achieving 89.5% accuracy on HAM10000 dataset. Improved lesion classification by 4.7% using VGG16 and custom CNNs. Devised Flask based diagnostic web app.
- Implemented TF-IDF, document categorization, and feature creation for automatic news text summarization.

Market Basket Analysis & Recommendation System | Apriori, Collaborative Filtering Oct 2022 - Nov 2022

- Optimized recommendation and marketing strategies using algorithms like Apriori, Item and User Collaborative Filtering, and customer segmentation via clustering using K-Means on a 200K user Instacart dataset.

ACHIEVMENT

Awards: Hack Harvard 2022([Project](#)) - **Most-Liked** project in Devpost; a video content summarizer within 36 hours.

EDUCATION

NORTHEASTERN UNIVERSITY

Boston, MA

Master of Science in Data Science

Jan 2022-Dec 2023

- Relevant Course: Supervised/ Unsupervised Learning, Natural Language Processing, Computer Vision, Cloud.

KIIT UNIVERSITY

Bhubaneswar, IND

Bachelor of Engineering, Major in Electronics Engineering

Jun 2014-Apr 2018